

A. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

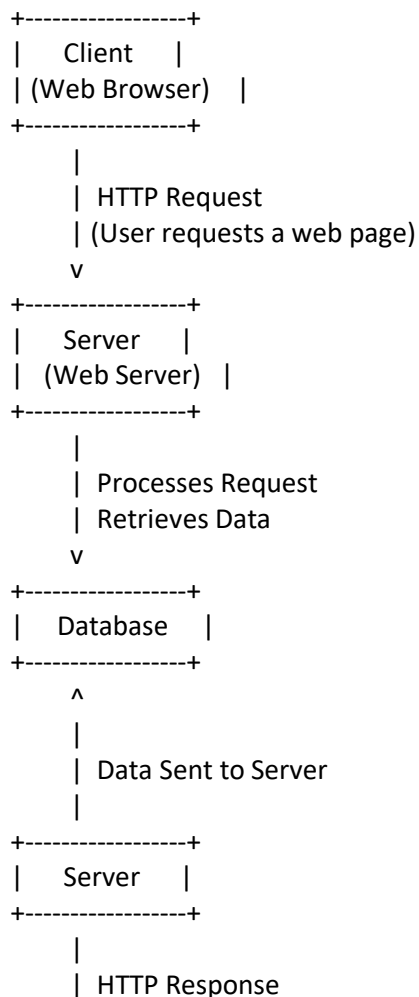
Ans-1. Frontend Development: Frontend development is the part of a website or application that users can see and interact with. It focuses on designing the user interface, such as buttons, menus, images, forms, and layouts. Frontend developers mainly use HTML, CSS, and JavaScript to build attractive and responsive web pages. Ex. Amazon product page design.

2. Backend Development: Backend development is the part of a website that works behind the scenes. It handles the server, database, and business logic. The backend processes user requests, stores data, and sends the required information back to the frontend. Common backend technologies include Java, Python, PHP, Node.js, MySQL, and MongoDB. Ex. Gmail login verification and data storage.

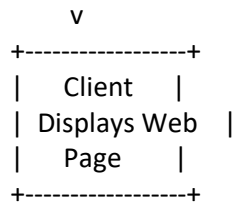
3. Full-Stack Development: Full-stack development combines both frontend and backend development. A full-stack developer can build the complete website or application, from designing the user interface to managing the server and database. Ex. Building a complete food delivery website from start to finish.

B. Create a simple diagram showing how the client-server model works in web architecture.

Ans- CLIENT – SERVER MODEL



| (HTML, CSS, JavaScript)



C. Describe how a browser requests and displays a web page from a web server.

Ans. A browser displays a web page by following a few simple steps:

- The user enters a website address (URL) into the browser or clicks a link.
- The browser sends an HTTP request to the web server asking for the requested web page.
- The web server receives the request, processes it, and collects the required files such as HTML, CSS, JavaScript, and images.
- The server sends these files back to the browser as an HTTP response.

D. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Ans.1. Visual Studio Code (VS Code): VS Code is a code editor where we write and edit our HTML, CSS, JavaScript, and other programming files. It is easy to use and supports many useful extensions.

2. Web Browser (Google Chrome or Mozilla Firefox): A web browser is used to open and test web pages. It also provides Developer Tools, which help us inspect the code and fix errors while developing a website.

3. Git: Git is a version control tool that keeps track of changes made to the project. It allows developers to save different versions of their code and work together without losing previous work.

4. GitHub: GitHub is an online platform where developers store and share their projects. It also makes collaboration easier by allowing multiple people to work on the same project.

E. Explain what a web server is and give examples of commonly used servers.

Ans. A web server is a software or computer system that stores website files and delivers them to users when they request a web page through a browser. When a user enters a website's URL, the browser sends a request to the web server. The server processes the request and sends back the required files, such as HTML, CSS, JavaScript, images, and other resources, so the browser can display the web page.

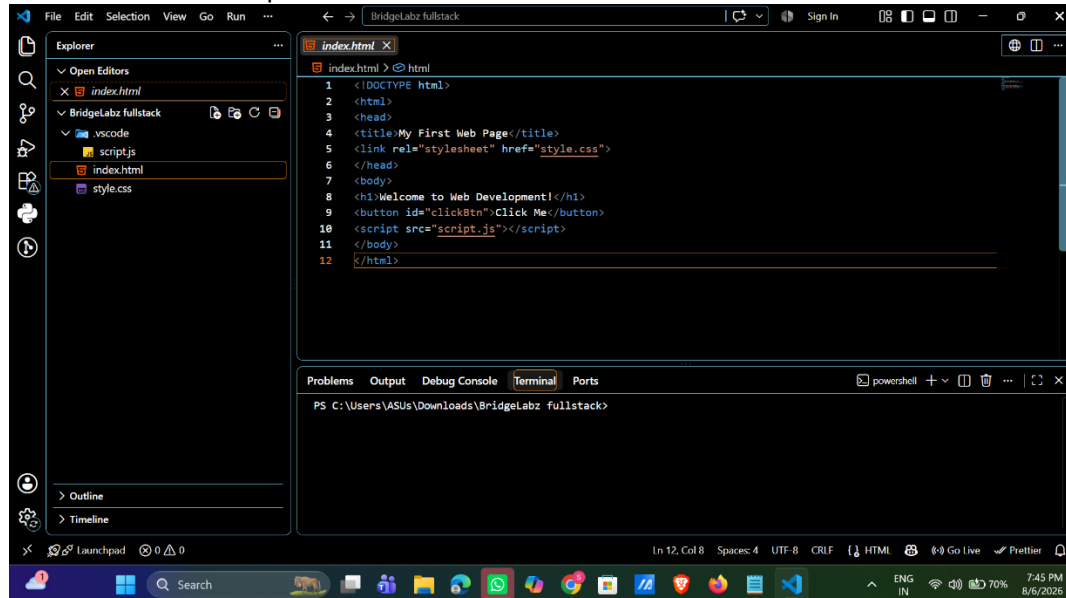
F. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans. Frontend Developer: A frontend developer creates the part of the website that users see and interact with. They use HTML, CSS, and JavaScript to design the user interface and improve the user experience.

Backend Developer: A backend developer works on the server side of the website. They handle the application's logic, process user requests, and connect the website to the database.

Database Administrator (DBA): A database administrator manages the database by storing, organizing, securing, and maintaining the data so that it can be accessed efficiently whenever needed.

G. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup .



H. Explain the difference between static and dynamic websites. Provide an example of each.

Ans. A static website displays the same content to every user. The web pages are pre-built using HTML and CSS, and the content only changes when a developer manually updates the files.

Example: A personal portfolio website or a company's "About Us" page.

A dynamic website displays content that can change based on the user, time, or data stored in a database. It is built using server-side technologies such as PHP, Java, Python, or Node.js.

Example: Facebook, Amazon, or an online shopping website where the content changes according to the user's activity.

Difference: A static website has fixed content and is simple to build, whereas a dynamic website generates content dynamically and provides interactive features for users.

I. Research and list five web browsers. Explain how rendering engines differ between them.

Ans. Different web browsers use different rendering engines to display web pages. A rendering engine is responsible for reading HTML, CSS, and JavaScript and converting them into the web page that users see on their screen.

1. Google Chrome – Uses the Blink rendering engine.
2. Mozilla Firefox – Uses the Gecko rendering engine.
3. Microsoft Edge – Uses the Blink rendering engine.
4. Apple Safari – Uses the WebKit rendering engine.
5. Opera – Uses the Blink rendering engine.

How Rendering Engines Differ:

Although all rendering engines display web pages, they process HTML, CSS, and JavaScript in slightly different ways. This can affect the speed, appearance, and performance of a website. As a result, a website may look or behave slightly differently across different browsers, which is why developers test their websites on multiple browsers.

J. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans. WEB ARCHITECTURE FLOW

